EE2C2 FPGA Lab — Report Worksheet
Combinational Logic, Sequential Logic, LED FSM, and PWM

Student names	______________________________
Student numbers	______________________________
Group size (2 / 4)	__________ Date: _______________
Board ID / bench	____________________

This worksheet follows the same topics as the lab handout. Work through the handout topic by topic and fill in the matching section here. *Paste evidence* boxes are for your own captures (waveforms, reports, the board). This lab is graded **pass/fail**; brief, correct answers are enough.

Topic 0 — Setup

Checkpoint 0. Confirm the project was created and record your Vivado version (2023.2).

☐ The AC701 Vivado project was created (`build/EE2C2_FPGA_Lab.xpr` exists).

Vivado version (should be 2023.2): ____________________

Topic 1 — Combinational logic

Checkpoint 1a. Which line is an AND gate, which is an XOR gate, and which is a multiplexer?

- Line that describes an **AND** gate: ______________________________
- Line that describes an **XOR** gate: ______________________________
- Line that describes a **multiplexer**: ______________________________

Simulation evidence: ☐ ran `tb_comb_led_logic`

Paste evidence: `tb_comb_led_logic` waveform — output changing at the same instant as an input.

(drop your screenshot / photo here)

Checkpoint 1b. Why can a testbench change inputs with `#10` delays, while synthesizable RTL must not use `#10` delays?

Topic 2 — Sequential logic

Checkpoint 2. What hardware does `always_ff @(posedge clk)` infer, and why is `tick_1s` a clock-enable pulse rather than a new clock?

Paste evidence: A zoom of `count_value` wrapping and the one-cycle `tick_1s` pulse.

(drop your screenshot / photo here)

Topic 3 — Finite state machines

Checkpoint 3. Using the FSM waveform, choose two consecutive `tick_1s` pulses after reset. For each pulse, record the state just before the pulse, the state just after the pulse, and the LED/RGB output after the transition.

`tick_1s` pulse	State before pulse	State after pulse	LED/RGB output after transition
1			
2			

Then explain what this shows about the roles of the state register, next-state logic, and output-decode logic.

Simulation evidence: ☐ ran `tb_led_fsm`

Paste evidence: `tb_led_fsm` waveform — reset region and two `tick_1s`-to-`state_dbg` transitions.

(drop your screenshot / photo here)

Topic 4 — Synthesis and implementation

- ☐ Synthesis completed with no errors.
- ☐ Implementation completed with no errors.
- ☐ AC701 bitstream generated (`.../impl_1/top_ac701.bit`).

Checkpoint 4. In one sentence, what is the difference between synthesis and implementation?

Topic 5 — Hardware targets and AUP-ZU3 connection

Checkpoint 5. Why do Topics 4, 6, and 7 use the AC701 project, while the AUP-ZU3 board demos use prebuilt bitstreams? In one sentence, also state why `top_ac701.bit` must not be programmed onto the AUP-ZU3.

Topic 6 — Static timing analysis

Quantity	Value
Clock period	
WNS (Worst Negative Slack)	
TNS (Total Negative Slack)	
WHS, if shown	
THS, if shown	
Failing endpoints, if shown	
Did timing meet? (yes / no)	

Critical (worst) path:

Field	Value
Startpoint	
Endpoint	
Clock domain / path group	
Data path delay, if shown	
Logic levels, if shown	

Checkpoint 6. Record WNS and TNS; state whether timing is met and how you know; record the critical path start/endpoint; and describe the path in one sentence.

Paste evidence: Timing summary and/or the critical-path schematic.

(drop your screenshot / photo here)

Topic 7 — Power analysis

Quantity	Value
Total on-chip power	
Static power	
Dynamic power	
Clock power	
Logic / signal power	
I/O power, if shown	
Largest category	

Checkpoint 7a. Which power category is largest, and does the $\alpha C V_{\mathrm{DD}}^2 f$ relation explain why?

Checkpoint 7b. What assumptions drive the estimate, and what would you provide to improve its confidence level?

Checkpoint 7c. Does this report include the power consumed by the physical LEDs on the board?

Paste evidence: The Power Summary panel.

(drop your screenshot / photo here)

Topic 8 — PWM DAC

☐ I completed the PWM DAC topic under `pwm_dac/`.

Responsible TA signature/initials after seeing `PWM_STUDENT_TEST_PASS`:

____

Checkpoint 8a. For a 100 MHz clock and 8-bit counter, what is the PWM frequency? Show the arithmetic.

Checkpoint 8b. Why does changing duty change LED brightness?

Checkpoint 8c. Which TODO line describes sequential logic, and which describes combinational logic?

Checkpoint 8d. Why does `sw[3:0]=1111` not produce a true 100% duty cycle in this selector?

Paste evidence: Console or `xsim.log` evidence showing `PWM_STUDENT_TEST_PASS`.

(drop your screenshot / photo here)